ENTERPRISE DATA GOVERNANCE FRAMEWORK DEVELOPMENT & ROADMAP FOR A GLOBAL MANUFACTURING COMPANY

DATA GOVERNANCE POLICY

Document Control

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1. POLICY OVERVIEW

1.1. Purpose

This Enterprise Data Governance Policy establishes the principles, governance structures, accountability, standards, controls, and processes required to manage enterprise data as a **strategic business asset** across the Global Manufacturing Company.

The policy is designed to ensure that enterprise data is:

- Accurate and reliable
- Complete and consistent
- Secure and appropriately protected
- Accessible to authorized users
- Clearly owned and governed
- Properly classified and managed throughout its lifecycle
- Traceable through appropriate metadata and lineage
- Compliant with applicable legal, regulatory, contractual, and corporate requirements
- Fit for operational, analytical, reporting, and strategic decision-making purposes

The policy establishes the foundation for a scalable Data Governance Operating Model supporting **global manufacturing operations, digital transformation, Industry 4.0, advanced analytics, and AI initiatives**.

1.2. Scope

This policy applies to:

Organizational Scope

- Corporate functions
- Manufacturing plants
- Regional business units
- Shared service centers
- Supply chain organizations
- Engineering organizations
- Procurement
- Finance
- Sales and Customer Management
- Quality
- Maintenance
- IT
- Data & Analytics teams

Data Scope

The policy covers:

- Master Data
- Transactional Data
- Operational Data
- Reference Data
- Analytical Data
- Metadata
- Historical Data
- Sensitive Data
- Personal Data
- Machine and IoT Data

System Scope

The policy applies to data managed by or exchanged through:

- ERP
- MES
- PLM
- WMS
- SCM
- CRM
- QMS
- EAM
- SCADA
- Industrial IoT platforms

- Data Warehouse
- Data Lake
- Cloud platforms
- Business Intelligence platforms
- Enterprise applications
- Third-party data platforms

2. POLICY OBJECTIVES

The Enterprise Data Governance Program shall pursue the following objectives:

2.1. Establish Enterprise Accountability

- Define clear ownership and accountability for enterprise data.
- Assign Data Owners and Data Stewards to critical data domains.
- Establish clear decision rights for data-related decisions.

2.2. Improve Data Quality

- Establish enterprise Data Quality standards.
- Define Critical Data Elements (CDEs).
- Implement measurable Data Quality Rules and thresholds.
- Reduce data duplication, inconsistency, and errors.

2.3. Establish a Single Source of Truth

- Standardize critical master data across enterprise systems.
- Establish Golden Records through Master Data Management (MDM).
- Minimize conflicting versions of business-critical data.

2.4. Strengthen Data Security and Privacy

- Protect Confidential and Restricted data.
- Enforce appropriate access controls.
- Apply encryption, masking, monitoring, and audit logging where required.

2.5. Improve Data Transparency

- Establish an Enterprise Data Catalog.
- Maintain a Business Glossary.
- Document data lineage from source through transformation to consumption.

2.6. Support Regulatory Compliance

- Align data management practices with applicable laws, regulations, contractual obligations, and corporate requirements.
- Maintain sufficient evidence to support audits and compliance assessments.

2.7. Enable Digital Transformation

Provide a trusted data foundation for:

- Industry 4.0
- Industrial IoT
- Advanced Analytics
- Artificial Intelligence
- Predictive Maintenance
- Supply Chain Analytics
- Digital Manufacturing
- Executive Business Intelligence

3. DATA GOVERNANCE PRINCIPLES

The Company shall adopt the following principles:

3.1. Data as a Strategic Asset

Enterprise data shall be managed as a strategic corporate asset rather than solely as an IT resource.

3.2. Business Ownership

Business functions shall own the meaning, quality, and appropriate use of their data domains.

3.3. Accountability

Every critical data domain and Critical Data Element shall have clearly assigned accountability.

3.4. Data Quality by Design

Data quality controls shall be embedded into business processes and systems rather than relying exclusively on downstream correction.

3.5. Single Source of Truth

Authoritative sources shall be defined for critical master and reference data.

3.6. Least Privilege

Access shall be granted based on business need and minimum required privileges.

3.7. Security and Privacy by Design

Security and privacy considerations shall be incorporated into data processes and solutions from the beginning.

3.8. Transparency and Traceability

Critical data should be discoverable, understandable, and traceable through appropriate metadata and lineage.

3.9. Lifecycle Management

Data shall be governed from creation through storage, usage, archival, retention, and secure disposal.

3.10. Continuous Improvement

Governance capabilities shall continuously evolve based on KPI performance, business requirements, technology changes, audit findings, and lessons learned.

4. ROLES & RESPONSIBILITIES (RACI-ALIGNED)

4.1. Executive Steering Committee

The Executive Steering Committee shall:

- Provide executive sponsorship
- Approve strategic governance direction
- Resolve enterprise-level conflicts
- Approve major investments
- Review material data risks

4.2. Data Governance Council

The Data Governance Council shall:

- Approve enterprise data standards
- Approve governance policies
- Resolve cross-domain data conflicts
- Review governance KPIs
- Prioritize major remediation initiatives
- Approve policy exceptions where appropriate

4.3. Chief Data Officer (CDO)

The CDO shall:

- Own the enterprise Data Governance strategy
- Chair or sponsor governance leadership
- Report governance performance to executive leadership
- Resolve strategic data issues
- Sponsor enterprise-wide data initiatives
- Drive data culture and adoption

4.4. Data Governance Office (DGO)

The DGO shall:

- Operate the governance framework
- Maintain policies and standards
- Coordinate Data Owners and Data Stewards
- Manage governance processes
- Monitor KPIs
- Coordinate issue escalation
- Maintain governance documentation
- Conduct maturity assessments

4.5. Data Owners

Data Owners are accountable for the business definition, quality, security requirements, and appropriate use of data within their domains.

Responsibilities include:

- Approving business definitions
- Establishing quality expectations
- Approving Data Quality Rules
- Approving access requirements
- Approving data standards
- Prioritizing data remediation
- Reviewing governance KPIs

4.6. Data Stewards

Data Stewards are responsible for operational governance.

Responsibilities include:

- Maintaining business metadata
- Monitoring data quality
- Investigating data issues
- Coordinating remediation
- Maintaining business glossary terms
- Supporting data classification
- Validating data changes

4.7. IT Data Custodians

IT Data Custodians are responsible for technical implementation.

Responsibilities include:

- System availability.
- Technical access controls.
- Data integration.
- Backup and recovery.
- Encryption.
- Technical metadata.
- System-level data controls.

RACI Overview

Governance Activity	DGC	CDO	DGO	DO	DS	IT	EA	SEC
Approve Data Governance Policy	A	R	C	I	I	I	I	C
Approve Enterprise Standards	A	R	C	C	I	I	C	C
Define Business Glossary	I	C	R	A	R	I	C	I
Define Critical Data Elements	I	C	R	A	R	I	I	I
Data Quality Rules	I	C	C	A	R	C	I	I
Master Data Governance	I	C	R	A	R	C	C	I
Metadata Management	I	C	A	R	R	C	C	I
Data Quality Monitoring	I	C	C	A	R	C	I	I
Issue Resolution	I	C	C	A	R	C	I	I
KPI Reporting	C	A	R	C	C	I	I	I
Access Approval	I	C	I	A	C	R	I	C
Security Controls	I	I	I	C	I	R	C	A

Legend: R = Responsible, A = Accountable, C = Consulted, I = Informed

Abbreviation	Full Name	Primary Responsibility
EA	Enterprise Architect	Data architecture, integration, technology standards, enterprise architecture alignment
SEC	Information Security	Access control, encryption, cybersecurity, security classification, security risk

5. DATA CLASSIFICATION POLICY

5.1. Classification Levels

Level	Description	Example
Public	Information approved for unrestricted external distribution. Disclosure is not expected to cause material harm to the Company, its employees, customers, suppliers, or business operations.	• Public website content • Published annual reports • Approved press releases • Public product brochures • Public sustainability reports • Publicly available job postings
Internal	General business information intended for authorized employees, contractors, and approved business partners. Unauthorized disclosure may cause limited operational or reputational impact.	• Internal procedures • Organizational information • Standard operating procedures • Non-sensitive operational reports • Internal training materials • General production schedules
Confidential	Sensitive business information that requires controlled access because unauthorized disclosure could cause significant financial, operational, competitive, contractual, or reputational impact.	• Supplier contracts • Customer contracts • Financial forecasts • Procurement pricing • Detailed production reports • Non-public sales data • Inventory information • Internal management reports
Restricted (PII / highly sensitive)	Highly sensitive information requiring the highest level of protection because unauthorized access, disclosure, alteration, or loss could result in significant legal, regulatory, financial, operational, security, privacy, or competitive impact.	• Customer/Employee Personally Identifiable Information (PII) • Payroll data • Credentials • Security information • Trade secrets • Proprietary product formulas • Sensitive engineering IP • Unreleased product designs • Sensitive M&A information • Security logs containing sensitive information

5.2. Handling Rules

Level	Encryption	Access	Masking
Public	Not mandatory unless required by the transmission/storage platform	Open access; external distribution permitted when formally approved	Not required
Internal	Recommended for Data in transit; required where mandated by corporate security standards	Employees and authorized contractors/business partners with legitimate business need	Generally not required; may be applied where appropriate
Confidential	**Required** for data in transit and at rest using approved corporate security controls	Restricted to authorized personnel based on business need and least privilege; RBAC required	Required for sensitive attributes where access does not require full values
Restricted	**Mandatory** encryption at rest and in transit using approved enterprise security standards	Strictly controlled; least privilege, RBAC and/or ABAC where appropriate; MFA required for sensitive systems; Data Owner approval required	**Mandatory** for PII and other sensitive attributes when full values are not required; tokenization or equivalent controls may be used

5.3. Additional Handling Requirements

1. **Public**
   - May be distributed externally when approved for public release.
   - Must not contain confidential business information or sensitive personal information.
   - Public release should follow applicable Corporate Communications procedures.
2. **Internal**
   - Access should be limited to authorized workforce members and approved third parties.
   - External sharing requires appropriate authorization.
   - Data should not be published externally without classification review.

3. Confidential

- Access shall follow the Least Privilege principle.
- External sharing requires Data Owner approval and appropriate contractual/security controls.
- Confidential information shall not be stored on unauthorized personal devices or applications.
- Access shall be periodically reviewed.
- Appropriate audit logging shall be implemented for critical systems.

4. Restricted

- Access shall require a clearly documented business need.
- Data Owner approval shall be required for sensitive access.
- MFA shall be mandatory for applicable systems.
- Data shall be encrypted during transmission and while stored.
- PII and other sensitive attributes shall be masked, tokenized, or otherwise protected when full values are unnecessary.
- Access shall be logged and monitored.
- Periodic access reviews shall be performed at least quarterly.
- Restricted data shall only be transferred through approved secure channels.
- Additional contractual and regulatory controls shall apply where required.

5.4. Manufacturing-Specific Classification Examples

Data Domain	Example Data	Typical Classification
Material Master	Material ID, Material Description	Internal
Product Master	Product Code, Product Category	Internal
Supplier Master	Supplier Name, Supplier ID	Internal / Confidential
Supplier Pricing	Contract Price, Negotiated Terms	Confidential
Customer Master	Customer ID, Customer Name	Internal / Confidential
Customer PII	Personal Email, Phone Number, Address	Restricted
Production Data	Production Volume, Production Schedule	Confidential
BOM	Bill of Materials	Confidential / Restricted
Engineering	Product Design, Technical Drawings	Confidential / Restricted
SCADA/IoT	Machine Telemetry, Sensor Data	Internal / Confidential
Equipment Data	Equipment ID, Maintenance Schedule	Internal

Data Domain	Example Data	Typical Classification
Financial Data	Budget, Forecast, Cost Data	Confidential
Employee Data	Employee ID, Payroll Information	Restricted
Security Data	Credentials, Security Keys	Restricted
Public Information	Published Product Brochure	Public

6. CRITICAL DATA ELEMENTS

Critical Data Elements shall be identified using the following criteria:

- Regulatory or audit impact
- Financial impact
- Production impact
- Customer impact
- Supply chain impact
- Quality or product compliance impact
- Operational dependency
- Cross-system dependency
- Executive reporting dependency
- Security or privacy impact

Example CDEs:

Domain	CDE
Material	Material ID
Product	Product Code
Manufacturing	Production Order
Engineering	BOM
Supplier	Supplier ID
Customer	Customer ID
Inventory	Inventory Quantity
Plant	Plant Code
Equipment	Equipment ID
Quality	Inspection Result
Finance	Cost Center

7. DATA QUALITY POLICY

7.1. Data Quality Dimensions

The Company shall establish enterprise Data Quality Management based on the following dimensions:

Dimension	Definition
Accuracy	Data correctly represents the real-world business object or event.
Completeness	Required data attributes are populated.
Consistency	Data is consistent across systems, processes, and business domains.
Validity	Data conforms to predefined formats, business rules, and reference values.
Uniqueness	Duplicate records are prevented or appropriately managed.
Timeliness	Data is available and updated within the required business timeframe.
Integrity	Relationships between related data entities remain valid and complete.

Minimum Requirements:

- Define Data Quality Rules for CDEs
- Establish quality thresholds
- Perform profiling
- Implement automated monitoring where feasible
- Track data quality issues
- Conduct Root Cause Analysis (RCA)
- Monitor remediation effectiveness

Target:
Overall Enterprise Data Quality Score: ≥95%

7.2. Data Quality Rules

The following rules represent illustrative enterprise Data Quality Rules for the manufacturing environment.

Data Quality Rules (Example)

Domain	Rule	Threshold
Material Master	Material ID must be unique	≥99.5%
Material Master	Material Description must not be null	≥99%
Material Master	Material Type must contain an approved reference value	100%
Material Master	Base Unit of Measure must use an approved value	100%
Material Master	Material status must be valid and current	≥99%
Product Master	Product Code must be unique	≥99.5%
Product Master	Product must have an assigned product category	≥99%
Supplier Master	Supplier ID must be unique	≥99.5%
Supplier Master	Required supplier attributes must be populated	≥98%
Supplier Master	Supplier status must use approved values	100%
Customer Master	Customer ID must be unique	≥99.5%
Customer Master	Mandatory customer attributes must be populated	≥98%
BOM	Every active product must have a valid BOM where applicable	100%
BOM	BOM component IDs must exist in Material Master	100%
BOM	BOM quantities must be greater than zero	100%
Production	Production Order must reference a valid Material/Product	100%
Production	Production Order must reference a valid Plant	100%
Production	Production quantities must be valid and non-negative	100%
Inventory	Inventory quantity must not contain invalid negative values unless explicitly permitted by business rules	≥99%

Domain	Rule	Threshold
Inventory	Inventory records must reference valid Material and Plant IDs	100%
Equipment	Equipment ID must be unique	≥99.5%
Equipment	Equipment must be assigned to a valid Plant	100%
Equipment	Equipment status must use approved reference values	100%
Quality	Inspection results must reference valid inspection orders	100%
Procurement	Purchase Order must reference a valid Supplier ID	100%
Finance	Cost Center must use an approved master value	100%
Integration	Critical records must reconcile between sources and target systems	≥99%
Metadata	Critical Data Elements must have an assigned Data Owner	100%

7.3. Data Quality Scorecard

The Data Quality Scorecard shall provide management with a standardized view of data quality performance across critical business domains.

Data Quality Scorecard — Example

Domain	Rule ID	KPI	Target
Material Master	DQ-MAT-001	Material ID Uniqueness	≥99.5%
Material Master	DQ-MAT-002	Mandatory Attribute Completeness	≥99%
Material Master	DQ-MAT-003	Reference Value Validity	100%
Product Master	DQ-PRD-001	Product Code Uniqueness	≥99.5%
Product Master	DQ-PRD-002	Product Attribute Completeness	≥99%
Supplier Master	DQ-SUP-001	Supplier ID Uniqueness	≥99.5%
Supplier Master	DQ-SUP-002	Supplier Master Completeness	≥98%
Customer Master	DQ-CUS-001	Customer ID Uniqueness	≥99.5%

Domain	Rule ID	KPI	Target
Customer Master	DQ-CUS-002	Customer Data Completeness	≥98%
BOM	DQ-BOM-001	BOM Completeness	100%
BOM	DQ-BOM-002	BOM Component Validity	100%
Production	DQ-PRD-001	Production Order Validity	≥99%
Inventory	DQ-INV-001	Inventory Data Accuracy	≥99%
Equipment	DQ-EQP-001	Equipment Master Accuracy	≥99%
Quality	DQ-QLT-001	Inspection Data Completeness	≥99%
Procurement	DQ-PUR-001	Supplier Reference Integrity	100%
Finance	DQ-FIN-001	Cost Center Validity	100%
Integration	DQ-INT-001	Cross-System Reconciliation	≥99%
Metadata	DQ-MET-001	CDE Ownership Coverage	100%

7.4. Enterprise Data Quality Score

An overall Data Quality Score shall be calculated using the performance of approved Data Quality Rules.

Example:

$$\textbf{Enterprise Data Quality Score} = \frac{\textbf{Σ Weighted DQ Rule Scores}}{\textbf{Σ Applicable Rule Weights}}$$

Critical Data Elements and business-critical domains may receive higher weighting than lower-risk data.

Target State:
Enterprise Data Quality Score ≥95%

Suggested interpretation:

Score	Status	Management Action
≥95%	🟢 Green	Target achieved; continue monitoring
90–94.9%	🟡 Amber	Investigation and remediation required
<90%	🔴 Red	Escalation and corrective action required

7.5. Data Quality Monitoring

1. Automated data quality checks
2. Data Quality Scorecard
3. Issue tracking & remediation workflow

Monitoring Principles
Data Quality shall be monitored continuously or at an appropriate frequency based on:

- Data criticality
- Business process frequency
- Operational impact
- Regulatory requirements
- Data volatility
- System capabilities

Critical operational data should be monitored **daily or near-real-time where technically feasible**, while less volatile reference and master data may be monitored weekly or monthly.

8. DATA ACCESS & SECURITY POLICY

8.1. Access Control

8.1.1. Role-Based Access Control (RBAC)

Role-Based Access Control (RBAC) shall be mandatory for enterprise applications and data platforms where technically feasible.
Access shall be assigned based on predefined business and technical roles rather than individually granting unrestricted access to data.

Example RBAC Model:

Role	Example Access
Production Operator	Assigned production and machine-operation data
Production Manager	Production KPIs, schedules, and operational reports
Supply Chain Analyst	Inventory, logistics, and supply chain data
Procurement Manager	Supplier and procurement information
Engineering User	Approved product, BOM, and engineering data
Finance User	Financial and cost information
Data Steward	Metadata, Data Quality, and governed data management functions

Role	Example Access
Data Owner	Data approval, access authorization, and governance functions
IT Administrator	Technical administration based on approved privileged access
Executive	Authorized management and enterprise reporting data

Access shall be aligned with the user's actual responsibilities.

8.1.2. Attribute-Based Access Control (ABAC)

Attribute-Based Access Control (ABAC) may be implemented as an advanced control mechanism for high-volume, geographically distributed, or context-sensitive environments.

ABAC decisions may consider:

- User role
- Business unit
- Manufacturing plant
- Country/region
- Data classification
- Device trust level
- Network/location context
- Employment status
- Time of access
- Application
- Data ownership
- Business purpose

Example:
A production manager may access production data only for:
Assigned Plant + Authorized Business Unit + Approved Data Classification
rather than receiving unrestricted access to production data across all global plants.

8.1.3. Least Privilege

The **Least Privilege Principle shall be mandatory**.
Users shall receive only the minimum level of access required to perform their approved responsibilities.

Controls shall include:

- No default access to sensitive data
- Access based on documented business requirements
- Removal of unnecessary permissions
- Periodic access recertification
- Immediate revocation when access is no longer required
- Separate privileged administrative accounts where appropriate

8.1.4. Segregation of Duties

Conflicting responsibilities shall be appropriately separated to reduce fraud, unauthorized changes, and operational risk.

Example:
A single individual should not normally have unrestricted ability to:
Create Supplier → Approve Supplier → Create Purchase Order → Approve Payment
unless an approved exception and compensating control exists.

8.2. Authentication & Authorization

8.2.1. Authentication Requirements

Multi-Factor Authentication (MFA)

Multi-Factor Authentication (MFA) shall be mandatory for sensitive systems and privileged access.

MFA should apply to systems containing:

- Restricted data
- Personal data
- Financial information
- Engineering IP
- Security information
- Production-critical systems
- Administrative functions
- Cloud platforms
- Data platforms

Examples include:

- ERP administrative functions
- Cloud data platforms
- Data Governance platforms
- MDM
- Data Catalog
- BI administration
- Privileged IT accounts

8.2.2. Privileged Access Management (PAM)

All privileged accounts shall be managed through a PAM solution.

Privileged accounts include:

- Database Administrators
- System Administrators
- Cloud Administrators
- Security Administrators

Requirements:

✓ Session Monitoring
✓ Password Vaulting
✓ Privileged Session Recording
✓ Just-In-Time (JIT) Access
✓ Temporary Privilege Elevation

8.3. Periodic Access Review

Access to enterprise data shall be reviewed **at least quarterly** for sensitive and critical systems.

Quarterly Access Review

The review shall validate:

- User identity
- Employment status
- Business role
- Current responsibilities
- Assigned permissions
- Data classification
- Business justification
- Segregation of Duties
- Privileged access
- Third-party access

Review Outcome

Each access entitlement shall be:

- Approved
- Modified
- Revoked
- Escalated for investigation

Terminated employees and users who change roles shall have inappropriate access revoked or modified promptly in accordance with the Company's identity and access management procedures.

8.4. User Lifecycle Management

8.4.1. Joiner

New employees receive access only after:

- Manager Approval
- Data Owner Approval
- Security Validation

8.4.2. Mover

Role changes trigger:

- Access Revalidation
- Privilege Adjustment
- SoD Review

8.4.3. Leaver

Upon termination:

- Immediate account deactivation
- Access revocation within 24 hours
- Token recovery
- Session termination

8.5. Data Protection

8.5.1. Data Encryption

8.5.1.1. Encryption at Rest

Data classified as **Confidential or Restricted** shall be protected using approved encryption mechanisms when stored.

Encryption shall apply, as appropriate, to:

- Databases
- Data Warehouses
- Data Lakes
- Cloud storage
- Backup repositories
- File storage
- Portable storage
- Critical application databases
- Encryption keys shall be managed using approved enterprise key-management practices.

Recommended Standards:

Control	Standard
Database Encryption	AES-256
File Encryption	AES-256
Backup Encryption	AES-256
Key Management	HSM or Enterprise KMS

8.5.1.2. Encryption in Transit

Confidential and Restricted data shall be protected during transmission using approved secure communication protocols.

This shall apply to data exchanged between:

- ERP ↔ MES
- ERP ↔ WMS
- ERP ↔ PLM
- MES ↔ SCADA
- SCADA ↔ IoT platforms
- Applications ↔ Data Warehouse

- Applications ↔ Data Lake
- Internal systems ↔ Cloud platforms
- Company ↔ Third-party systems
- Unencrypted transmission of sensitive data over untrusted networks shall be prohibited.

8.5.2. Data Masking

Data masking shall be required for Personally Identifiable Information (PII) when users do not require access to the complete underlying value.

Examples include:

PII Attribute	**Example Masking**
Mobile Number	+62-812-****-7890
Email Address	f****@company.com
National ID	**********1234
Bank Account	********5678
Employee ID	EMP-****-123
Home Address	Partial address displayed
Customer Name	F***** B*****

Masking shall be applied particularly to:

- Development environments
- Testing environments
- Analytics environments where full PII is unnecessary
- Non-production databases
- Self-service reporting
- Data extracts shared with authorized users

Where appropriate, **tokenization, pseudonymization, or anonymization** may be used instead of conventional masking.

8.5.3. Dynamic Data Masking

The system shall automatically determine masking levels based on user roles.

Example:

Role	View PII
Customer Service	Partial Masking
Relationship Manager	Partial Masking
Data Analyst	Fully Masked
Fraud Investigator	Full Access
Compliance Officer	Full Access

8.5.4. Data Loss Prevention (DLP)

The manufacturing company shall implement DLP controls to prevent unauthorized disclosure of sensitive data.

Monitoring Scope:

- Email
- Web Uploads
- USB Devices
- Cloud Storage
- Printing Activities
- Collaboration Platform

8.5.5. Security Monitoring & Audit Logging

All access to Confidential and Restricted Data shall be logged.

Audit logs must capture:

- User ID
- Timestamp
- Source IP Address
- Accessed Dataset
- Activity Type
- Approval Reference

Log retention period:
Minimum 1 year, or longer if required by banking regulations.

8.5.6. Data Backup & Recovery

Requirements:

- Daily Incremental Backup
- Weekly Full Backup
- Offsite Backup Storage
- Disaster Recovery Replication

Recovery Targets:

Metric	Target
Recovery Time Objective (RTO)	≤ 4 Hours
Recovery Point Objective (RPO)	≤ 15 Minutes

8.6. Manufacturing-Specific Security Controls

The Company's manufacturing environment requires additional consideration because operational technology may directly affect production.

Critical Systems:

- MES
- SCADA
- Industrial IoT
- PLC-related systems
- Production control systems
- EAM
- Quality systems

Access to production-critical systems shall be:

- Strictly controlled.
- Based on authorized roles.
- Logged and monitored.
- Subject to periodic review.
- Separated between operational and administrative responsibilities where appropriate.

Production systems shall not be exposed unnecessarily to public or untrusted networks.

8.7. Third-Party Access

Third-party access shall require:

- Business justification
- Contractual authorization
- Appropriate security requirements
- Defined access scope
- Time-bound access where feasible
- Data Owner approval
- Security approval for sensitive environments
- Periodic access review

Third parties shall receive only the minimum data and system access necessary to perform their contracted responsibilities.

8.8. Audit Logging & Monitoring

Critical systems shall maintain appropriate audit logs covering, where applicable:

- User authentication
- Access to sensitive data
- Data creation
- Data modification
- Data deletion
- Privileged activities
- Permission changes
- Security events
- Administrative actions

Logs shall be protected against unauthorized modification and retained according to applicable corporate and regulatory requirements.

8.9. Data Access Monitoring

Security and Data Governance teams should monitor for:

- Unusual access patterns
- Excessive data downloads
- Repeated failed authentication
- Unauthorized privilege escalation
- Access outside normal business patterns
- Access to data outside a user's business domain
- Suspicious third-party activity

Potential security or privacy incidents shall be handled according to the Company's **Data Security & Privacy Incident Management** process.

8.10. Access Exception Management

Exceptions to this policy shall:

1. Be formally documented
2. Provide a valid business justification
3. Identify affected data and systems
4. Assess associated risks
5. Define compensating controls
6. Specify an expiration date
7. Receive appropriate approval

Permanent exceptions should be avoided wherever technically and operationally feasible.

8.11. Key Security & Access KPIs

KPI	**Target**
Quarterly Access Review Completion	100%
Restricted Data Access with Approved Business Justification	100%
Sensitive System MFA Coverage	100%
Terminated User Access Revocation	100% within defined SLA
Critical Privileged Accounts Reviewed	100%
Confidential/Restricted Data Encrypted at Rest	100%
Confidential/Restricted Data Encrypted in Transit	100%
PII Masking in Non-Production Environments	100% where applicable
Unauthorized Access Incidents	0 material incidents
Critical Access Control Findings	0

9. DATA LIFECYCLE MANAGEMENT

9.1. Lifecycle Principles

The Company shall manage data according to the following principles:

1. Business Purpose
   Data shall be collected, created, and retained for a defined and legitimate business purpose.

2. Data Minimization
   Only data necessary for the intended business purpose should be collected and retained.

3. Security by Design
   Security controls shall be applied throughout the entire data lifecycle based on data classification and risk.

4. Retention Accountability
   Each data domain shall have defined retention requirements and an accountable Data Owner.

5. Regulatory Compliance
   Data retention and disposal shall comply with applicable legal, regulatory, tax, accounting, contractual, privacy, and industry requirements.

6. Controlled Disposal
   Data that has reached the end of its approved retention period shall be securely deleted or destroyed unless a legal hold or other approved exception applies.

9.2. Lifecycle Stages

- Creation
- Storage
- Usage
- Archival
- Deletion

9.2.1. Creation

Data may be created, collected, generated, or acquired through:

- Business applications
- Manufacturing systems
- Machine and IoT sensors
- Manual data entry
- Business transactions
- Supplier integrations
- Customer interactions

- Engineering processes
- Quality inspections
- Financial processes
- External data providers

Creation Controls

At the point of creation or acquisition:

- Data shall have an identified business purpose
- Appropriate Data Owners shall be assigned
- Data shall be classified according to the Data Classification Policy
- Required metadata shall be captured
- Data quality rules shall be applied to Critical Data Elements
- Personal data shall be collected and processed according to applicable privacy requirements
- Duplicate or unnecessary data collection should be avoided

Manufacturing Example

A production event generated by an MES may include:

- Production Order
- Material ID
- Plant ID
- Machine ID
- Production Quantity
- Timestamp
- Operator ID
- Quality Result

These data elements should be appropriately classified, governed, and linked to relevant metadata.

9.2.2. Storage

Data shall be stored in approved enterprise systems and repositories.

Examples include:

- ERP databases
- MES databases
- PLM repositories
- Data Warehouses
- Data Lakes
- Cloud storage
- IoT platforms
- Document repositories
- Backup systems

Storage Controls

Storage controls shall include:

- Access control based on RBAC/ABAC where applicable
- Encryption at rest for Confidential and Restricted data
- Appropriate backup and recovery controls
- Data quality monitoring for critical datasets
- Metadata registration
- Appropriate retention tagging
- Protection against unauthorized modification or deletion

Data shall not be stored indefinitely without an approved business, legal, regulatory, or operational justification.

9.2.3. Usage

Data may be accessed and processed for authorized purposes, including:

- Manufacturing operations
- Production planning
- Supply chain management
- Inventory management
- Engineering
- Quality management
- Maintenance
- Financial reporting
- Customer management
- Business intelligence
- Analytics
- Artificial intelligence and machine learning

Usage Controls

Users shall:

- Access data only for authorized business purposes
- Follow the Least Privilege principle
- Comply with the applicable Data Classification Policy
- Not copy sensitive data unnecessarily
- Not transfer data through unauthorized channels
- Protect PII and other Restricted information
- Follow applicable data quality and security requirements

9.2.4. Archival

Data that is no longer actively required but must be retained for business, legal, regulatory, historical, or operational purposes shall be moved to an approved archive.

Examples

Manufacturing data that may be archived includes:

- Completed production orders
- Historical inventory records
- Historical purchase orders
- Closed supplier records
- Historical quality inspection records
- Historical maintenance records
- Obsolete product information
- Historical financial records
- Historical engineering documentation

Archival Controls

Archived data shall:

- Maintain its original classification
- Remain protected against unauthorized access
- Retain required metadata
- Remain searchable/retrievable where required
- Maintain data integrity
- Have a defined retention expiration date
- Be subject to applicable legal holds

Archiving shall not be treated as an exemption from security or privacy requirements.

9.2.5. Deletion

Data shall be securely deleted when:

- The approved retention period has expired
- The data is no longer required for a legitimate business purpose
- There is no applicable legal or regulatory retention requirement
- No legal hold or investigation requires preservation
- The Data Owner has approved disposal where required

Deletion shall include, where applicable:

- Production databases
- File repositories
- Cloud storage

- Data Warehouse/Lake environments
- Archive systems
- Replicated datasets
- Temporary processing environments

9.3. Retention Policy

The following retention periods represent illustrative baseline examples for the manufacturing case study.

Data Type	**Example Retention**
Production Transaction Data	5 years
Purchase / Procurement Transactions	7 years
Sales Transactions	7 years
Financial / Accounting Data	7 years
Customer Data	Active + 2 years
Supplier Data	Active + 3 years
Employee Data	Employment + applicable statutory period
Inventory Data	5 years
Quality Inspection Data	5–10 years
Maintenance / Equipment History	5–10 years
Engineering / Product Records	Product lifecycle + applicable retention period
BOM / Product Master History	Product lifecycle + applicable retention period
Manufacturing Logs	1–3 years
Security Logs	1 year minimum
Application Logs	1 year
IoT / Machine Telemetry	1–3 years, depending on business requirements
Audit Records	7 years or applicable requirement
Backup Data	According to approved backup retention schedule

Note: *Retention periods shown are illustrative defaults for the case study. Actual retention periods must be confirmed by* ***Legal, Compliance, Privacy, Finance, Records Management, and relevant Data Owners*** *based on applicable laws, regulations, tax/accounting requirements, contractual obligations, product liability requirements, industry standards, and country-specific requirements.*

9.4. Secure Deletion

Secure deletion shall be mandatory when data reaches the end of its approved lifecycle and is not subject to a legal hold or other approved exception.

Deletion controls shall be appropriate to the storage technology and sensitivity of the data.

Secure Deletion Principles

- Data shall be deleted using approved enterprise mechanisms
- Restricted data requires enhanced deletion controls
- Deletion from primary systems shall consider applicable replicas and managed copies
- Where complete deletion is technically impractical, compensating controls shall be documented and approved
- Deletion activities shall be auditable
- Destruction of physical media shall use approved secure disposal methods

9.5. Backup and Replication Considerations

Backup and disaster-recovery copies shall be managed according to approved retention schedules.

The Company shall define:

- Backup retention periods
- Recovery-point objectives (RPO)
- Recovery-time objectives (RTO)
- Backup encryption requirements
- Access controls
- Restoration procedures
- Secure disposal procedures

Expired backup copies shall be securely disposed of according to the approved backup retention schedule.

9.6. Data Lifecycle Monitoring

The Data Governance Office shall monitor compliance with lifecycle requirements.

Key Monitoring Metrics

KPI	Target
Critical Data with Assigned Retention Period	100%
CDEs with Defined Lifecycle Stage	100%
Retention Policy Compliance	≥95%
Unauthorized Data Retention	0 material instances
Overdue Data Disposal	<5%
Restricted Data Secure Deletion Compliance	100%
Legal Holds Properly Implemented	100%
Data Owners Assigned to Critical Domains	100%

10. METADATA & DATA CATALOG POLICY

10.1. Metadata Management Principles

The Company shall manage metadata according to the following principles:

1. Metadata as an Enterprise Asset
   Metadata shall be treated as a governed enterprise asset that enables users to understand and trust organizational data.

2. Business Meaning First
   Critical data shall have clear and standardized business definitions that are understandable to business users.

3. Ownership and Accountability
   Critical data assets shall have an assigned Data Owner and Data Steward.

4. Traceability
   Critical data should be traceable from its source through transformation and integration to its final consumption.

5. Classification
   Data assets shall have an approved classification aligned with the Company's Data Classification Policy.

6. Single Enterprise View
   The Data Catalog shall provide a consistent enterprise view of governed data assets across business domains and manufacturing locations.

7. Continuous Maintenance
   Metadata shall be reviewed and updated when data structures, business definitions, ownership, systems, or processes change.

10.2. Requirements

1. All critical data must be registered in the data catalog
2. Business glossary must be maintained

10.3. Metadata Coverage

1. Data definitions
2. Data owner & steward
3. Lineage
4. Data classification

11. DATA LINEAGE & TRACEABILITY

1. End-to-end lineage must be documented
   - Target: 100% of Tier-1 / critical datasets have documented end-to-end lineage within the data catalog.
2. Impact analysis required before changes
3. Mandatory for critical datasets

12. MASTER DATA MANAGEMENT (MDM)

12.1. Scope

MDM shall provide a trusted source of master data for critical business processes and systems, including:

- ERP
- MES
- PLM
- WMS
- SCM
- QMS
- EAM
- CRM
- Data Warehouse
- Data Lake
- BI and Analytics platforms

The MDM scope shall initially prioritize the following domains.

- Material Master Data
- Product Master Data
- Bill of Materials (BOM)
- Supplier Master Data
- Customer Master Data
- Plant / Location Master Data
- Equipment / Asset Master Data
- Organizational Master Data

12.2. Rules

12.2.1. Single Source of Truth

Each Master Data Domain shall have one authoritative source recognized as the official System of Record (SoR).

12.2.2. Golden record management

The MDM solution shall maintain a **Golden Record** representing the most trusted and governed version of a master-data entity.

A Golden Record should contain:

- Unique enterprise identifier
- Standardized attributes
- Validated business values
- Source-system references
- Data quality status
- Effective dates
- Lifecycle status
- Ownership information

12.2.3. Unique Identification

Each master entity shall have a unique enterprise identifier wherever feasible.

Examples:

- Material ID
- Product ID
- Supplier ID
- Customer ID
- Equipment ID
- Plant ID
- Location ID

Duplicate identifiers shall not be permitted within the authoritative master-data repository.

12.2.4. Standardization

Master data shall follow standardized naming, coding, formatting, and reference-data conventions.

Examples:

- Unit of Measure

  Use standardized values:

  - KG
  - EA
  - L
  - M
  - MM

  rather than uncontrolled variations such as:

  - Kilo
  - Kilogram
  - Kg.
  - KGS

- Country

  Use approved country codes rather than free-text variations.

12.2.5. Deduplication

The Company shall implement systematic duplicate detection and remediation processes.

Duplicate detection may use:

- Exact matching
- Fuzzy matching
- Business rules
- Reference data
- Cross-system identifiers
- Address matching
- Tax identifiers
- Supplier registration numbers

Example:

Supplier A:
ABC Industrial Ltd.

Supplier B:
ABC Industrial Limited

Supplier C:
ABC INDUSTRIAL LTD

The MDM process should identify these as potential duplicates and route them for validation.

12.2.6. Synchronization

Approved master data shall be distributed to consuming systems through controlled integration mechanisms.

Examples:

- APIs
- ETL/ELT pipelines
- Event-driven integration
- Batch interfaces
- Enterprise integration platforms

Synchronization failures shall be monitored and managed through Data Issue Management.

12.3. MDM Data Quality Rules

MDM data shall be subject to defined quality rules covering:

- Completeness
- Accuracy
- Consistency
- Uniqueness
- Validity
- Timeliness
- Conformity

12.3.1. MDM Data Quality Rules — Example

Domain	Data Quality Rule	KPI / Threshold
Material	Material ID must be unique	100%
Material	Material description must not be blank	≥99%
Material	Valid Unit of Measure required	100%
Material	Material type must use approved values	100%
Material	Active material must have valid plant assignment	≥99%
Product	Product ID must be unique	100%
Product	Product lifecycle status must be valid	100%
BOM	Component must exist in Material Master	100%
BOM	BOM quantity must be greater than zero	100%
BOM	BOM version must be unique per product/plant	100%
Supplier	Supplier ID must be unique	100%
Supplier	Supplier name must not be blank	≥99%
Supplier	Tax/registration identifier must be valid where required	≥99%
Supplier	Duplicate suppliers must be identified	≥98% detection
Customer	Customer ID must be unique	100%
Customer	Required customer attributes must be complete	≥98%

Domain	Data Quality Rule	KPI / Threshold
Customer	Country must use approved reference values	100%
Plant	Plant code must be unique	100%
Location	Location must belong to a valid plant/business entity	100%
Equipment	Equipment ID must be unique	100%
Equipment	Serial number must be unique where applicable	100%
Organization	Cost Center must map to valid organizational hierarchy	100%

12.3.2 MDM Data Quality Scorecard

Dimension	KPI	Target
Completeness	Required master attributes populated	≥98%
Uniqueness	Duplicate master records	<1%
Validity	Values comply with approved reference data	≥99%
Consistency	Master data consistent across systems	≥98%
Accuracy	Validated against authoritative source	≥98%
Timeliness	Approved changes synchronized within SLA	≥95%
Conformity	Data follows enterprise standards	≥98%
Overall MDM Quality Score	Weighted DQ score	≥95%

12.3.3. MDM Issue Management

MDM data-quality issues shall follow the enterprise Data Issue Management process:
Detect → Log → Classify → Assign → Investigate → Remediate → Validate → Close

Examples include:

- Duplicate material
- Duplicate supplier
- Invalid UOM
- Missing product classification
- Incorrect plant assignment
- Inconsistent equipment identifier
- Invalid BOM component
- Customer master duplication
- Failed synchronization

Critical MDM issues affecting production, procurement, financial reporting, or regulatory obligations shall be escalated according to the approved Escalation Framework.

13. DATA ISSUE MANAGEMENT

13.1. Issue Classification

- Critical (business impact)
- High
- Medium
- Low

SLA Table (Example):

Severity	Initial Response	Target Resolution
Critical	< 4 Hours	< 24 Hours
High	< 1 Business Day	< 5 Business Days
Medium	< 2 Business Days	< 10 Business Days
Low	< 5 Business Days	< 30 Business Days

13.2. Workflow

1. Issue detected
2. Logged in tracking system
3. Assigned to Data Steward
4. Root cause analysis
5. Resolution & validation

13.3. Data Security & Privacy Incident Management

1. Detect and classify incidents
2. Containment and access revocation where necessary
3. Notify Security, Privacy Officer, Legal, and Governance Lead
4. For regulated personal-data incidents, follow applicable notification obligations
5. Post-incident RCA and corrective actions

14. COMPLIANCE & REGULATORY

14.1. Applicable Regulations

14.1.1. Personal Data Protection

The Company shall comply with applicable personal-data protection and electronic-system regulations in the jurisdictions where it operates, collects, stores, or processes personal data.

For operations involving Indonesia, the framework shall consider:

- **Indonesia Personal Data Protection Law (UU No. 27/2022)**.
- Relevant implementing regulations and sector-specific requirements applicable to the Company's operations.
- Applicable electronic-system and cybersecurity requirements.

Where appropriate, **GDPR principles may be adopted as a reference framework**, particularly for global operations or cross-border data processing, subject to Legal and Privacy assessment.

14.2.2. Cross-Border Data Considerations

Because the Company operates globally, data may be transferred between:

- Headquarters
- Regional offices
- Manufacturing plants
- Shared service centers
- Cloud environments
- Third-party service providers

Cross-border transfers of personal or regulated data shall be assessed against applicable local requirements.

The Company shall consider:

- Data residency requirements.
- Cross-border transfer restrictions.
- Appropriate contractual safeguards.
- Vendor/third-party requirements.
- Privacy impact assessments where applicable.
- Security controls.
- Data classification.
- Legal and regulatory requirements of both originating and receiving jurisdictions.

14.2.3. Financial Transaction Compliance

Financial and transaction-related data shall be governed according to applicable:

- Tax requirements
- Accounting standards
- Financial reporting requirements
- Payment regulations
- Corporate record-retention requirements
- Anti-fraud requirements
- Anti-money-laundering requirements where applicable
- Audit requirements
- Country-specific financial regulations

Examples of potentially regulated or critical financial data include:

- General Ledger transactions
- Accounts Payable
- Accounts Receivable
- Purchase Orders
- Supplier invoices
- Customer invoices
- Payment records
- Tax records
- Cost accounting
- Product costing
- Financial reporting data

Financial records shall maintain appropriate **accuracy, integrity, traceability, retention, and auditability**.

14.2. Audit

- Regular internal audit
- External audit readiness

14.3. Audit Trail

- Critical governance activities shall maintain appropriate audit trails
- Audit logs shall be protected against unauthorized modification and retained according to applicable requirements

14.4. Non-Compliance

- Material or repeated non-compliance shall be escalated according to severity
- Not every issue requires escalation to the Governance Council; escalation shall be **risk- and severity-based**
- Corrective action required

14.5. Exception Management & Policy Waiver

Temporary exceptions to Data Governance Policies may be granted only when there is a legitimate business or technical justification and the associated risk is understood and controlled.

Exception requests shall be:

1. Documented
2. Risk-assessed
3. Approved by the Data Owner
4. Reviewed by Information Security where security implications exist
5. Reviewed by Legal/Compliance where regulatory implications exist
6. Reviewed by the Governance Council for high-risk cases
7. Time-bound
8. Supported by compensating controls
9. Supported by a remediation plan

14.6. Exception Expiration

Every temporary exception shall include a defined expiration date.

Before expiration, the exception shall be:

- Closed because remediation is complete
- Renewed with documented justification and reapproval
- Escalated if the risk has increased
- Rejected if continued exception is no longer acceptable

High-risk exceptions should receive enhanced monitoring and periodic review.

15. GOVERNANCE KPIs

The Governance KPI framework establishes measurable indicators to monitor the **effectiveness, adoption, compliance, and business value** of the Enterprise Data Governance Framework.

The KPI framework covers five major dimensions:

1. Data Quality
2. Ownership & Stewardship
3. Data Issue Management
4. Security & Privacy
5. Master Data & Classification

The Data Governance Office (DGO) shall consolidate KPI results and provide periodic reporting to the **Data Governance Council and executive management**.

Governance KPI Dashboard

KPI	Definition	Target	Measure-ment Frequency	Owner
Data Quality Score	Weighted score across defined Data Quality dimensions for critical manufacturing and enterprise data	≥95%	Monthly	Data Steward
Data Ownership Coverage	% of critical data assets with formally assigned and approved Data Owners	100%	Monthly	DGO
Data Stewardship Coverage	% of critical data assets with assigned Data Stewards	100%	Monthly	DGO
Data Incident Reduction	YoY reduction in material/ critical data incidents	≥20% reduction YoY	Quarterly	DGO
Open Critical Data Incidents	Number of unresolved Critical data incidents past defined escalation criteria	0	Weekly / Monthly	Data Owner
Repeat Data Issues	% of data issues recurring after previous remediation	<5%	Monthly	Data Steward
Critical Issue Resolution SLA Compliance	% of Critical issues resolved within approved SLA	≥95%	Monthly	Data Owner

KPI	Definition	Target	Measure-ment Frequency	Owner
Overall SLA Compliance	% of all governed data issues resolved within their assigned SLA	≥95%	Monthly	DGO
Customer PII Protection Compliance	% of applicable Customer PII assets meeting required access, classification, masking, encryption, and protection controls	100%	Monthly / Quarterly	Privacy / Security
Unauthorized Data Access Incidents	Confirmed incidents involving unauthorized access to governed enterprise data	0	Monthly	Information Security
Data Classification Coverage	% of critical data assets with approved classification	100%	Monthly	Data Steward
Master Data Accuracy	% of sampled master records validated as accurate against authoritative sources	≥99%	Monthly	MDM / Data Owner

16. ENFORCEMENT

1. Mandatory compliance for all employees
2. Violations subject to disciplinary action

17. REVIEW & MAINTENANCE

1. Policy reviewed annually
2. Updated based on regulatory changes